Name: S.Rithrishareddy
Batch: COMETFWC028
Date: 10 July 2025

GATE 2009, ECE Question Number 38

Abstract

This project simulates a vending machine logic circuit using an Arduino UNO and a 7-segment display, based on GATE 2009 Question 59. The logic uses two inputs (P1 and P2) to determine product selection and display values (0, 2, 5, or E). This implementation enhances understanding of combinational logic, input handling, and segment control in embedded systems.

1. Components

Component	Qty
Arduino UNO	1
Push Buttons (P1, P2)	2
7-Segment Display (Common Cathode)	1
220Ω Resistors	7
Breadboard	1
Jumper Wires	12
Laptop with Arduino IDE	1

Table 1: List of components used

2. Setup and Connections

- Connect push button P1 to D2 and P2 to D3 with pull-down resistors.
- Connect 7-segment display segments:
  - a → D4, b → D5, c → D6, d → D7, e → D8, f → D9, g → D10
- Connect each segment pin through a 220Ω resistor.
- Common cathode pin of display to GND.
- Arduino GND connected to breadboard GND.

3. Logic Summary

- Inputs: P1, P2 (1 = pressed, 0 = not pressed)
- Segment g: $g = P1 + P2$
- Segment e: $e = b + c$
- Segment d: $d = c + e$
- Display outputs: 0, 2, 5, or E depending on input

4. Pin Mapping

- **P1** – D2 (Input)
- **P2** – D3 (Input)
- **7-Segment Display:**
  - a – D4, b – D5, c – D6, d – D7
  - e – D8, f – D9, g – D10

5. Analysis

5.1 Truth Table

P1	**P2**	**Display**
0	0	0
1	0	2
0	1	5
1	1	E

5.2 Segment Activation

Digit	a	b	c	d	e	f	g
0	1	1	1	1	1	1	0
2	1	1	0	1	1	0	1
5	1	0	1	1	0	1	1
E	1	0	0	1	1	1	1

5.3 Derivations

- $g = P1 + P2$
- $e = b + c$
- $d = c + e$

6. GitHub Repository

- Repository with code and schematics:
  `https://github.com/Rithrishareddy/RITHRISHA_FWC/tree/main/Hardware`

7. Conclusion

This hardware implementation using an Arduino and a 7-segment display successfully simulates the vending machine logic as asked in GATE 2009. The use of logical expressions and real-time button inputs enables verification of combinational logic through physical observation. Such exercises deepen understanding of digital systems and embedded hardware control.

8. Learning Outcomes

- Understood the functioning of 7-segment display and interfacing with Arduino.
- Gained hands-on experience in implementing logical expressions using hardware.
- Learned how to apply theoretical concepts to solve real-world exam problems.
- Practiced pin configuration, code uploading, and input/output validation.